Morning Paper Report - Friday, August 07, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Efficient Allocation of Working Memory Resource for Utility Maximization in Humans and Recurrent Neural Networks

Qingqing Yang, Hsin-Hung Li
Advances in Neural Information Processing Systems 38, 2025

Source: Crossref; matched terms: working memory, neural network, neural networks, recurrent neural network

Link: https://doi.org/10.52202/085713-1928

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.

2. Visual Working Memory Load Modulates Distraction in Visual Search

Jennifer Trujillo, Bo Yeong Won
Journal of Vision, 2025

Source: Crossref; matched terms: visual working memory, visual search, working memory

Link: https://doi.org/10.1167/jov.25.9.2278

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3. The AI Hippocampus: How Far are We From Human Memory?

Zixia Jia, Jiaqi Li, Yipeng Kang, Yuxuan Wang, Tong Wu, Quansen Wang, et al.
arXiv.org, 2026

Source: Semantic Scholar; matched terms: fidelity, transformer, human memory

Link: https://www.semanticscholar.org/paper/c4ca30f139c8b5b64e4dab53b450471ffd777ffe

Goal: Memory plays a foundational role in augmenting the reasoning, adaptability, and contextual fidelity of modern Large Language Models and Multi-Modal LLMs

Method: Memory plays a foundational role in augmenting the reasoning, adaptability, and contextual fidelity of modern Large Language Models and Multi-Modal LLMs

Results: Memory plays a foundational role in augmenting the reasoning, adaptability, and contextual fidelity of modern Large Language Models and Multi-Modal LLMs

Conclusion: Extending beyond text, the survey examines the integration of memory within multi-modal settings, where coherence across vision, language, audio, and action modalities is essential Key architectural advances, benchmark tasks, and open challenges are discussed, including issues related to memory capacity, alignment, factual consistency, and cross-system interoperability.

4. Learning the Synaptic and Intrinsic Membrane Dynamics Underlying Working Memory in Spiking Neural Network Models

Yinghao Li, Robert Kim, Terrence J. Sejnowski
Neural Computation, 2021

Source: Crossref; matched terms: working memory, neural network, recurrent neural network

Link: https://doi.org/10.1162/neco_a_01409

Goal: Abstract Recurrent neural network (RNN) models trained to perform cognitive tasks are a useful computational tool for understanding how cortical circuits execute complex computations

Method: Abstract Recurrent neural network (RNN) models trained to perform cognitive tasks are a useful computational tool for understanding how cortical circuits execute complex computations

Results: Training our model on a wide range of cognitive tasks resulted in diverse yet task-specific synaptic and membrane parameters

Conclusion: Further dissecting the optimized parameters revealed that fast membrane properties are important for encoding stimuli, and slow synaptic dynamics are needed for WM maintenance This approach offers a unique window into how connectivity patterns and intrinsic neuronal properties contribute to complex dynamics in neural populations.

5. Activated long-term memory and visual working memory during hybrid visual search: Effects on target memory search and distractor memory

Stephanie M. Saltzmann, Brandon Eich, Katherine C. Moen, Melissa R. Beck
Memory & Cognition, 2024

Source: Crossref; matched terms: visual working memory, visual search, working memory

Link: https://doi.org/10.3758/s13421-024-01556-1

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